Free pythagorean theorem worksheet worksheet 18

Maths | 8th grade | Mixed Operations

1. Evaluate: (3 + 2) × (5 - 3) ÷ (4 ÷ 2)

A) 2
B) 4
C) 5
D) 6

2. If x = 3, what is the value of 2x^2 - 4x + 1?

A) 11
B) 7
C) 5
D) 9

3. Which of the following is equal to 3/4 + 2/3?

A) 17/12
B) 13/12
C) 19/12
D) 5/7

4. Solve for y: 2(y + 3) = 6y - 4

A) 2
B) 3
C) 4
D) 1

5. What is the value of (8 - 3^2) + 4?

A) 5
B) -1
C) 3
D) 7

6. A car travels 60 miles in 1.5 hours. What is the average speed in miles per hour?

A) 30
B) 40
C) 50
D) 45

7. Simplify: 5[2(3 + 1) - 4] + 6

A) 22
B) 30
C) 18
D) 26

8. If 3x = 18, what is 6/x?

A) 1
B) 2

C) 3
D) 4

9. Calculate: 7 - 2 × 3 + 4 ÷ 2

A) 6
B) 5
C) 7
D) 8

10. Solve: 15 ÷ (5 - 3) × 2 + 7

A) 22
B) 23
C) 25
D) 30

Answer Key

1. D
First, solve inside the parentheses: (3 + 2) = 5 and (5 - 3) = 2. Then, (4 ÷ 2) = 2. Now, multiply and divide: 5 × 2 ÷ 2 = 10 ÷ 2 = 5.

2. B
Substitute x with 3: 2(3)^2 - 4(3) + 1 = 2(9) - 12 + 1 = 18 - 12 + 1 = 7.

3. A
Find a common denominator (12): (3/4 = 9/12) + (2/3 = 8/12) = 17/12.

4. C
Expand and rearrange: 2y + 6 = 6y - 4. Move terms: 6 + 4 = 6y - 2y. Thus, 10 = 4y, so y = 2.5.

5. B
Calculate exponent first: 3^2 = 9. Then, 8 - 9 + 4 = -1 + 4 = 3.

6. D
Speed = Distance/Time = 60 miles / 1.5 hours = 40 mph.

7. A
Inside the brackets first: 2(3 + 1) = 8, 8 - 4 = 4. Multiply and add: 5 × 4 = 20, then 20 + 6 = 26.

8. D
Find x first: x = 18/3 = 6. Substitute: 6/x = 6/6 = 1.

9. C
Order of operations: 7 - 6 + 2 = 1 + 2 = 3.

10. B
Evaluate inside the parentheses first: (5 - 3) = 2, then 15 ÷ 2 = 7.5, multiply by 2 = 15, then add 7 = 22.